

1 Sign-Mixing Edge Lemma (Conditional)

Lemma (Conditional). Assume $\sum_v (6 - d(v)) = 0$ and there exists at least one edge of degree 3. Then there exists an edge $e = (u, v)$ of degree 3 such that

$$(d(u) - 6)(d(v) - 6) < 0.$$

Proof (Averaging Argument). Define $K(v) = 6 - d(v)$ and partition vertices into $S^+ = \{v : K(v) > 0\}$ and $S^- = \{v : K(v) < 0\}$. Then

$$\sum_{v \in S^+} K(v) = \sum_{v \in S^-} (-K(v)) > 0.$$

Assume no sign-mixing edge exists among edges of degree 3. Then all such edges lie entirely within S^+ , S^- , or S^0 .

Let $W_3(v)$ denote the number of incident edges of degree 3. Then

$$\sum_{v \in S^+} W_3(v) = 2|E_3(S^+)|, \quad \sum_{v \in S^-} W_3(v) = 2|E_3(S^-)|.$$

Define

$$\Sigma_3^+ = \sum_{v \in S^+} K(v)W_3(v), \quad \Sigma_3^- = \sum_{v \in S^-} (-K(v))W_3(v).$$

Averaging over edges of degree 3 gives

$$\Sigma_3^+ - \Sigma_3^- = \sum_{e=(u,v) \in E_3} (K(u) + K(v)).$$

Under the no-mixing assumption, this forces $\Sigma_3^+ \cdot \Sigma_3^- = 0$, contradicting curvature balance and nonempty E_3 .

Hence a sign-mixing edge exists.